

ORF 524 — Final Exam

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Notes:

- Duration of examination: 180 minutes.
An additional 15 minutes are given to upload your solutions to Gradescope.
- All parts are equally weighted (5 points).
Please provide as much detail as possible in your answers.
Answers without proper justification will not receive full credit.
- Please start each question on a separate page.
A single PDF file must be submitted to Gradescope.
- Only one page of personal notes are allowed during the examination.
This is otherwise a closed-book and individual examination.

1 MLE and FGLS Poisson Regression (30 Points)

Suppose $(y_i, \mathbf{x}_i')$, $i = 1, 2, \dots, n$, is a random sample with y_i a scalar outcome and $\mathbf{x}_i$ a random vector of regressors supported on a compact set $\mathcal{X} \subset \mathbb{R}^d$, satisfying $y_i | \mathbf{x}_i \sim \text{Poisson}(\mathbf{x}_i' \boldsymbol{\beta}_0)$ with $\boldsymbol{\beta}_0 \in \mathcal{B}$ a vector of unknown parameters, where $\mathcal{B}$ is an open subset of $\mathbb{R}^d$. It is assumed that $\inf_{\mathbf{x} \in \mathcal{X}, \boldsymbol{\beta} \in \mathcal{B}} \mathbf{x}' \boldsymbol{\beta} > 0$.

1. Derive the average (scaled by $1/n$) conditional (on $\mathbf{x}_i$) log-likelihood, denoted by $\ell_n(\boldsymbol{\beta})$.

Derive the (conditional) score function, denoted by $\ell_{n,\boldsymbol{\beta}}(\boldsymbol{\beta})$.

Derive the (conditional) Hessian function, denoted by $\ell_{n,\boldsymbol{\beta}\boldsymbol{\beta}}(\boldsymbol{\beta})$.

Solution: The log-likelihood is

$$\ell_n(\boldsymbol{\beta}) = \frac{1}{n} \log \prod_{i=1}^n \frac{e^{-\mathbf{x}_i' \boldsymbol{\beta}} (\mathbf{x}_i' \boldsymbol{\beta})^{y_i}}{y_i!} = \frac{1}{n} \sum_{i=1}^n \{-\mathbf{x}_i' \boldsymbol{\beta} + y_i \log(\mathbf{x}_i' \boldsymbol{\beta}) - \log y_i!\}.$$

The score and Hessian are

$$\ell_{n,\boldsymbol{\beta}}(\boldsymbol{\beta}) = \frac{\partial}{\partial \boldsymbol{\beta}} \ell_n(\boldsymbol{\beta}) = \frac{1}{n} \sum_{i=1}^n \left\{ -1 + \frac{y_i}{\mathbf{x}_i' \boldsymbol{\beta}} \right\} \mathbf{x}_i,$$

and

$$\ell_{n,\boldsymbol{\beta}\boldsymbol{\beta}}(\boldsymbol{\beta}) = \frac{\partial}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \ell_n(\boldsymbol{\beta}) = -\frac{1}{n} \sum_{i=1}^n \frac{y_i}{(\mathbf{x}_i' \boldsymbol{\beta})^2} \mathbf{x}_i \mathbf{x}_i'.$$

2. Show that the first-order condition for the MLE estimator $\hat{\boldsymbol{\beta}}_{\text{ML}}$ can be written in the form:

$$\mathbf{0} = \frac{1}{n} \sum_{i=1}^n u_i(\hat{\boldsymbol{\beta}}) \mathbf{x}_i,$$

for some “pseudo-residual” function $u_i(\boldsymbol{\beta})$.

Show that $\mathbb{E}[u_i(\boldsymbol{\beta}_0) | \mathbf{x}_i] = 0$ with $\boldsymbol{\beta}_0$ denoting the population parameter value.

Solution: From the calculations above, and solving for the MLE first order condition,

$$u_i(\boldsymbol{\beta}) = -1 + \frac{y_i}{\mathbf{x}_i' \boldsymbol{\beta}},$$

and note that

$$\mathbb{E}[u_i(\boldsymbol{\beta}) | \mathbf{x}_i] = -1 + \frac{\mathbb{E}[y_i | \mathbf{x}_i]}{\mathbf{x}_i' \boldsymbol{\beta}} = -1 + \frac{\mathbf{x}_i' \boldsymbol{\beta}_0}{\mathbf{x}_i' \boldsymbol{\beta}} = 0. \quad \Leftrightarrow \quad \boldsymbol{\beta} = \boldsymbol{\beta}_0,$$

Therefore, $\mathbb{E}[u_i(\boldsymbol{\beta}_0) | \mathbf{x}_i] = 0$ as stated.

3. Show that $\sqrt{n} \ell_{n,\boldsymbol{\beta}}(\boldsymbol{\beta}_0) \rightsquigarrow \text{N}(\mathbf{0}, \mathcal{I}(\boldsymbol{\beta}_0))$.

Characterize the form of $\mathcal{I}(\beta_0)$.

Solution: By the standard CLT, because $\{u_i(\beta_0)\mathbf{x}_i : i = 1, 2, \dots, n\}$ are i.i.d. with $\mathbb{E}[u_i(\beta_0)\mathbf{x}_i] = 0$,

$$\sqrt{n}\ell_{n,\beta}(\beta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n u_i(\beta_0)\mathbf{x}_i \rightsquigarrow \mathbf{N}(\mathbf{0}, \mathcal{I}(\beta_0))$$

with

$$\mathcal{I}(\beta_0) = \mathbb{E} \left[\frac{y_i}{(\mathbf{x}_i' \beta_0)^2} \mathbf{x}_i \mathbf{x}_i' \right] = \mathbb{E} \left[\frac{1}{\mathbf{x}_i' \beta_0} \mathbf{x}_i \mathbf{x}_i' \right],$$

which gives the result.

4. Assume (here and below) that $\mathcal{I}(\beta_0)$ is strictly positive definite. Show that $\sqrt{n}(\hat{\beta}_{\text{ML}} - \beta_0) \rightsquigarrow \mathbf{N}(\mathbf{0}, \mathcal{V}(\beta_0))$ for some matrix $\mathcal{V}(\beta_0)$.

Characterize the form of $\mathcal{V}(\beta_0)$.

Show that the estimator $\hat{\beta}_{\text{ML}}$ is asymptotically efficient.

Solution: By standard M-estimation and MLE arguments (i.e., linearization of first-order conditions around β_0 , etc.), $\sqrt{n}(\hat{\beta}_{\text{ML}} - \beta_0) \rightsquigarrow \mathbf{N}(\mathbf{0}, \mathcal{V}(\beta_0))$ with $\mathcal{V}(\beta_0) = \mathcal{I}(\beta_0)^{-1}$. Finally, $\text{CRLB}[\beta_0] = \mathcal{I}(\beta_0)^{-1}$, which implies that $\hat{\beta}_{\text{ML}}$ is asymptotically efficient.

5. Since $\mathbb{E}[y_i|\mathbf{x}_i] = \mathbf{x}_i' \beta_0$ is linear, but $\mathbb{V}[y_i|\mathbf{x}_i] = \sigma_0^2(\mathbf{x}_i)$ is not constant, Generalized Least Squares (GLS) can be used to estimate β_0 more efficiently. Describe the steps to obtain an efficient feasible GLS estimator for this problem (exploiting the known form of the conditional variance), and derive an expression for the asymptotic covariance matrix of your estimator, assuming the Feasible GLS (FGLS) has the same asymptotic distribution as infeasible GLS. Compare the asymptotic distribution of the FGLS estimator to the MLE for this problem.

Solution: A FGLS (here, just WLS) estimator may be constructed as follows.

- (a) Use classical LS to regress y_i on x_i , obtaining the LS coefficient estimator $\hat{\beta}_{\text{LS}}$.
(b) Since $\mathbb{V}[y_i|\mathbf{x}_i] = \mathbf{x}_i' \beta_0$, create the “reweighted” observations

$$\tilde{y}_i = \frac{y_i}{\sqrt{\mathbf{x}_i' \hat{\beta}_{\text{LS}}}} \quad \text{and} \quad \tilde{\mathbf{x}}_i = \frac{\mathbf{x}_i}{\sqrt{\mathbf{x}_i' \hat{\beta}_{\text{LS}}}},$$

and regress $\tilde{y}_i$ on $\tilde{\mathbf{x}}_i$ using LS to obtain $\hat{\beta}_{\text{FGLS}}$.

- (c) Assuming $\hat{\beta}_{\text{FGLS}}$ has the same distribution as $\hat{\beta}_{\text{GLS}}$ (i.e., using the true β_0 instead of $\hat{\beta}_{\text{LS}}$), its asymptotic distribution will be

$$\sqrt{n}(\hat{\beta}_{\text{GLS}} - \beta_0) \rightsquigarrow \mathbf{N}(\mathbf{0}, \mathbf{V}_0^{-1}),$$

where, under standard regularity conditions,

$$\begin{aligned}\frac{\tilde{\mathbf{X}}'\tilde{\mathbf{X}}}{n} &= \frac{1}{n} \sum_{i=1}^n \frac{1}{\mathbf{x}_i'\hat{\boldsymbol{\beta}}_{\text{LS}}} \mathbf{x}_i \mathbf{x}_i' = \frac{1}{n} \sum_{i=1}^n \frac{1}{\mathbb{V}[y_i|\mathbf{x}_i]} \mathbf{x}_i \mathbf{x}_i' + o_{\mathbb{P}}(1) \\ &\rightarrow_{\mathbb{P}} \mathbf{V}_0 = \mathbb{E} \left[\frac{1}{\mathbf{x}_i'\boldsymbol{\beta}_0} \mathbf{x}_i \mathbf{x}_i' \right] = \mathcal{I}(\boldsymbol{\beta}_0).\end{aligned}$$

Therefore, the FWLS estimator has the same asymptotic distribution as the MLE, and is thus asymptotically efficient.

6. Suppose that $\mathbf{x}_i = (1, x_{i2}, x_{i3}, \dots, x_{id})'$ and you want to test the null hypothesis that the intercept term is one and the slope coefficients are zero, i.e., $\mathbf{H}_0 : \boldsymbol{\beta} = (1, 0, \dots, 0)'$. Derive the form of the feasible Lagrange multiplier (LM) or Score test statistic for this null hypothesis. Be as explicit as possible about the form of the estimator. Characterize the critical region of this testing procedure when $k = \dim(\boldsymbol{\beta}_0) = 5$ and the size of the test is $\alpha = 5\%$.

Solution: The “conditional” estimator of the information matrix, as a function of $\boldsymbol{\beta}$, is

$$\hat{\mathcal{I}}(\boldsymbol{\beta}) = \frac{1}{n} \sum_{i=1}^n \frac{1}{\mathbf{x}_i'\boldsymbol{\beta}} \mathbf{x}_i \mathbf{x}_i',$$

which should be evaluated at the restricted “estimator” $\bar{\boldsymbol{\beta}} = (1, 0, 0, \dots, 0)'$ for the LM statistic. Under $\mathbf{H}_0$, $\mathbf{x}_i'\bar{\boldsymbol{\beta}} = 1$, which simplifies many terms when evaluated at the restricted estimator. The Score (LM) test statistic is

$$\begin{aligned}T_{\text{LM}} &= n [\ell_{n,\boldsymbol{\beta}}(\bar{\boldsymbol{\beta}})]' [\hat{\mathcal{I}}(\bar{\boldsymbol{\beta}})]^{-1} [\ell_{n,\boldsymbol{\beta}}(\bar{\boldsymbol{\beta}})] \\ &= n \left[\frac{1}{n} \sum_{i=1}^n \left\{ -1 + \frac{y_i}{\mathbf{x}_i'\bar{\boldsymbol{\beta}}} \right\} \mathbf{x}_i' \right] \left[\frac{1}{n} \sum_{i=1}^n \frac{1}{\mathbf{x}_i'\bar{\boldsymbol{\beta}}} \mathbf{x}_i \mathbf{x}_i' \right]^{-1} \left[\frac{1}{n} \sum_{i=1}^n \left\{ -1 + \frac{y_i}{\mathbf{x}_i'\bar{\boldsymbol{\beta}}} \right\} \mathbf{x}_i \right] \\ &= n \left[\frac{1}{n} \sum_{i=1}^n \{-1 + y_i\} \mathbf{x}_i' \right] \left[\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \right]^{-1} \left[\frac{1}{n} \sum_{i=1}^n \{-1 + y_i\} \mathbf{x}_i \right].\end{aligned}$$

A Score test of $\mathbf{H}_0$ would reject if T_{LM} exceeded the upper α critical value of a χ_k^2 distribution. For the case of $k = 5$ and $\alpha = 5\%$, we would reject if the test statistic exceeded $\chi_5^2(0.05) = 11.07$.

2 Kernel Density Estimation (35 points)

Consider the following two definitions governing the dependence between random variables.

- A sequence of random variables $\{Y_i\}_{i=1}^\infty$ is called *stationary* if

$$\mathbb{P}[(Y_a, Y_{a+1}, \dots, Y_{b-1}, Y_b) \in A] = \mathbb{P}[(Y_{a+k}, Y_{a+1+k}, \dots, Y_{b-1+k}, Y_{b+k}) \in A]$$

for all positive integers $b \geq a$ and k , and all measurable sets A .

- A sequence of square-integrable random variables $\{Y_i\}_{i=1}^\infty$ is called *covariance stationary* if there is a number $\mu \in \mathbb{R}$ and a function $\gamma(\cdot)$ such that $\mathbb{E}[Y_i] = \mu$ for all i and $\text{Cov}[Y_i, Y_j] = \gamma(i - j)$ for all i, j .

Let $\{X_i\}_{i=1}^\infty$ be a stationary sequence of (real-valued) square-integrable random variables.

1. Show that $\{X_i\}_{i=1}^\infty$ is covariance stationary.

Solution: Stationarity implies that $(X_i : 1 \leq i \leq n)$ are identically distributed by setting $a = b = 1$, which implies $\mathbb{E}[X_i] = \mu$ for all i . Similarly, stationarity implies that $(X_i, X_j) \sim (X_{i+k}, X_{j+k})$ for all compatible i, j, k , and hence $\text{Cov}[X_i, X_j]$ only depends on $i - j$.

2. For $\bar{X} = \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$, show that $\bar{X} = \mathbb{E}[\bar{X}] + O_{\mathbb{P}}((\mathbb{V}[\bar{X}])^{1/2})$, where

$$\mathbb{E}[\bar{X}] = \mathbb{E}[X_i], \quad \mathbb{V}[\bar{X}] = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \gamma(i - j) = \frac{\gamma(0)}{n} + \frac{2}{n^2} \sum_{k=1}^{n-1} (n - k) \gamma(k).$$

Solution: By Markov inequality, for any $M > 0$, $\mathbb{P}[|W| > M] \leq \mathbb{E}[|W|^2]/M^2$, which means that if $\mathbb{E}[|W|^2] = O(1)$ then $|W| = O_{\mathbb{P}}(1)$. Setting $W = (\mathbb{V}[\bar{X}])^{-1/2}(\bar{X} - \mathbb{E}[\bar{X}])$, and noting that $\mathbb{E}[|W|^2] = 1$, it follows that $\bar{X} = \mathbb{E}[\bar{X}] + O_{\mathbb{P}}((\mathbb{V}[\bar{X}])^{1/2})$. By (covariance) stationarity, $\mathbb{E}[\bar{X}] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \mathbb{E}[X_i]$ because the random variables are $(X_i : 1 \leq i \leq n)$ equally distributed. Finally,

$$\mathbb{V}[\bar{X}] = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(X_i, X_j) = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \gamma(i - j) = \frac{\gamma(0)}{n} + \frac{2}{n^2} \sum_{k=1}^{n-1} (n - k) \gamma(k),$$

where the second equality follows from (covariance) stationarity (see first part), and the third equality follows by change of variables.

3. Show that if $\sum_{k=1}^\infty |\text{Cov}(X_i, X_{i+k})| < \infty$, then $\bar{X} \rightarrow_{\mathbb{P}} \mathbb{E}[X_i]$.

Solution: It suffices to show that $\mathbb{V}[\bar{X}] = o(1)$. Using the previous results,

$$\mathbb{V}[\bar{X}] \leq \frac{\gamma(0)}{n} + \frac{2}{n} \sum_{k=1}^{n-1} \frac{n - k}{n} |\gamma(k)| \leq \frac{\gamma(0)}{n} + \frac{1}{n} \sum_{k=1}^{n-1} |\gamma(k)| = O(n^{-1}),$$

under the assumptions imposed.

Consider the kernel density estimator based on the stationary sequence $\{X_i\}_{i=1}^\infty$:

$$\hat{f}(x) = \frac{1}{n} \sum_{i=1}^n w_n(X_i, x), \quad w_n(X_i, x) = \frac{1}{h} K\left(\frac{X_i - x}{h}\right),$$

where $K: \mathbb{R} \rightarrow \mathbb{R}$ is a P -th order kernel (where P is a positive integer), i.e. it is such that $u \mapsto u^j K(u)$ for $j = 0, 1, \dots, P$ are integrable and satisfy

$$\int_{\mathbb{R}} K(u) du = 1; \quad \int_{\mathbb{R}} u^j K(u) du = 0 \text{ for } j = 1, \dots, P-1; \quad \int_{\mathbb{R}} u^P K(u) du \neq 0,$$

and $h = h_n \rightarrow 0$ is a positive bandwidth. Assume also that $u \mapsto u^{P+1} K(u)$, $u \mapsto u K(u)^2$, $u \mapsto K(u)^2$ are integrable. Let $f(x)$ be the Lebesgue density of X_i , which is assumed to be positive, $P+1$ times continuously differentiable with $f^{(j)}(\cdot)$ bounded on $\mathbb{R}$ for all $j = 0, 1, \dots, P+1$.

4. Show that

$$\mathbb{E}[\hat{f}(x)] = f(x) + h^P f^{(P)}(x) B_P + O(h^{P+1}), \quad B_P = \frac{1}{P!} \int_{\mathbb{R}} u^P K(u) du.$$

Solution: Because the data is (covariance) stationary sequence, we have

$$\mathbb{E}[\hat{f}(x)] = \frac{1}{n} \sum_{i=1}^n \frac{1}{h} \mathbb{E}\left[K\left(\frac{X_i - x}{h}\right)\right] = \frac{1}{h} \mathbb{E}\left[K\left(\frac{X_i - x}{h}\right)\right] = \int_{\mathbb{R}} K(u) f(x + uh) du$$

and the result follows.

5. Assuming (here and below) that $\sum_{k=1}^\infty |\mathbb{C}ov(w_n(X_i, x), w_n(X_{i+k}, x))| < \infty$, show that

$$\mathbb{V}[\hat{f}(x)] = \frac{1}{nh} f(x) \|K\|^2 + O(n^{-1}), \quad \|K\|^2 := \int_{\mathbb{R}} K(u)^2 du.$$

Solution: Using the previous results,

$$\begin{aligned} \mathbb{V}[\hat{f}(x)] &= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \mathbb{C}ov(w_n(X_i, x), w_n(X_j, x)) \\ &= \frac{1}{n} \mathbb{V}(w_n(X_i, x)) + \frac{2}{n^2} \sum_{k=1}^n (n-k) \mathbb{C}ov[w_n(X_i, x), w_n(X_{i+k}, x)] \\ &= \frac{1}{nh} f(x) \int_{\mathbb{R}} K(u)^2 du + O(n^{-1}) \end{aligned}$$

because $\mathbb{V}(w_n(X_i, x)) = \mathbb{E}(w_n(X_i, x)) - (\mathbb{E}(w_n(X_i, x)))^2 = h^{-1} f(x) \int_{\mathbb{R}} K(u)^2 du + O(1)$.

6. Fix $x \in \mathbb{R}$. Find an expansion of the form

$$\mathbb{E}[(\hat{f}(x) - f(x))^2] = \text{AMSE}_n(x, h) + O(h^{2P+1} + n^{-1}),$$

where the leading term $\text{AMSE}_n(x, h)$ is some elementary function of h , $f^{(P)}(x)$, $f(x)$ (involving n and constants B_P , $\|K\|^2$).

Assuming $f^{(P)}(x) \neq 0$, find the optimal choice of bandwidth h to minimize the asymptotic mean square error of $\hat{f}(x)$, that is, find h minimizing $\text{AMSE}_n(x, h)$.

Solution: The assumptions and calculations above give

$$\mathbb{E}[(\hat{f}(x) - f(x))^2] = h^{2P}(f^{(P)}(x))^2 B_P^2 + \frac{1}{nh} f(x) \|K\|^2 + O(h^{2P+1} + n^{-1}),$$

which implies that the (approximate) MSE-optimal choice of bandwidth is

$$h = \left(\frac{f(x) \|K\|^2}{2P(f^{(P)}(x))^2 B_P^2 n} \right)^{1/(1+2P)}.$$

7. Show that

$$\int_{\mathbb{R}} |\hat{f}(x) - f(x)| f(x) dx = O_{\mathbb{P}} \left(h^P + \frac{1}{\sqrt{nh}} \right).$$

Solution: Using previous calculations, $\mathbb{E}[|\hat{f}(x) - f(x)|] = O(h^P + (nh)^{-1/2})$ uniformly in $x \in \mathbb{R}$. Thus, by Markov Inequality, and Fubini's theorem,

$$\begin{aligned} \mathbb{P} \left[\int_{\mathbb{R}} |\hat{f}(x) - f(x)| f(x) dx > M \right] &= \frac{1}{M} \mathbb{E} \left[\int_{\mathbb{R}} |\hat{f}(x) - f(x)| f(x) dx \right] \\ &= \frac{1}{M} \int_{\mathbb{R}} \mathbb{E} [|\hat{f}(x) - f(x)|] f(x) dx \\ &= \frac{1}{M} O(h^P + (nh)^{-1/2}), \end{aligned}$$

and the result follows.

3 Consistency of M-Estimators (35 Points)

Let $M_n : \Theta \mapsto \mathbb{R}$ be a sequence random functions ($n \geq 1$), and $M : \Theta \mapsto \mathbb{R}$ be a non-random function, satisfying for all $\epsilon > 0$:

$$\sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| = o_{\mathbb{P}}(1) \quad \text{and} \quad \sup_{\theta: d(\theta, \theta_0) \geq \epsilon} M(\theta) < M(\theta_0),$$

where limits are taken as $n \rightarrow \infty$, and $d : \Theta \times \Theta \mapsto \mathbb{R}_+$ denotes a distance function. Let $\hat{\theta}_n$ be a sequence of estimators ($n \geq 1$) satisfying $M_n(\hat{\theta}_n) \geq M_n(\theta_0) - o_{\mathbb{P}}(1)$.

1. Show that for all $\epsilon > 0$ there exists $\zeta > 0$ such that $\mathbb{P}[d(\hat{\theta}_n, \theta_0) \geq \epsilon] \leq \mathbb{P}[M(\hat{\theta}_n) < M(\theta_0) - \zeta]$.

Solution: By the second condition, which is an identification condition guaranteeing a well-separated local optima, for every $\epsilon > 0$ there exists $\zeta > 0$ such that $M(\theta) < M(\theta_0) - \zeta$ for every $\theta \in \Theta$ with $d(\theta, \theta_0) \geq \epsilon$. Thus, $\{d(\hat{\theta}_n, \theta_0) \geq \epsilon\} \subseteq \{M(\hat{\theta}_n) < M(\theta_0) - \zeta\}$, which gives

$$\mathbb{P}[d(\hat{\theta}_n, \theta_0) \geq \epsilon] \leq \mathbb{P}[M(\theta_0) - M(\hat{\theta}_n) > \zeta].$$

2. Show that $d(\hat{\theta}_n, \theta_0) = o_{\mathbb{P}}(1)$.

Solution: Using the previous result, it suffices to show that $M(\theta_0) - M(\hat{\theta}_n) \leq o_{\mathbb{P}}(1)$. Since $M_n(\theta_0) \xrightarrow{\mathbb{P}} M(\theta_0)$, we have $M_n(\hat{\theta}_n) \geq M_n(\theta_0) - o_{\mathbb{P}}(1) = M(\theta_0) - o_{\mathbb{P}}(1)$. This gives

$$M(\theta_0) - M(\hat{\theta}_n) \leq M_n(\hat{\theta}_n) - M(\hat{\theta}_n) + o_{\mathbb{P}}(1) \leq \sup_{\theta} |M_n(\theta) - M(\theta)| + o_{\mathbb{P}}(1) \xrightarrow{\mathbb{P}} 0,$$

where the last convergence is by the uniform convergence in probability assumption.

Consider the following example. For $(\mathbf{z}_i : i = 1, \dots, n)$ i.i.d., an inversely weighted M-estimator is:

$$\hat{\theta}_n = \arg \min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n \frac{\rho(\mathbf{z}_i; \theta)}{\hat{\omega}_i} \quad \text{and} \quad \theta_0 = \arg \min_{\theta \in \Theta} \mathbb{E} \left[\frac{\rho(\mathbf{z}_i; \theta)}{\omega_i} \right],$$

where the random sequence $(\hat{\omega}_i : 1 \leq i \leq n)$ satisfies

$$\max_{1 \leq i \leq n} |\hat{\omega}_i - \omega_i| = o_{\mathbb{P}}(1) \quad \text{and} \quad \inf_{1 \leq i \leq n} |\omega_i| \geq c > 0$$

for non-random constant c , $\theta \mapsto \mathbb{E} \left[\frac{\rho(\mathbf{z}_i; \theta)}{\omega_i} \right]$ is strictly convex, and $\mathbb{E}[\sup_{\theta \in \Theta} |\rho(\mathbf{z}_i; \theta)|^2] < \infty$. Furthermore, assume that

$$\sup_{\theta \in \Theta} \left| \frac{1}{n} \sum_{i=1}^n \frac{\rho(\mathbf{z}_i; \theta)}{\omega_i} - \mathbb{E} \left[\frac{\rho(\mathbf{z}_i; \theta)}{\omega_i} \right] \right| = o_{\mathbb{P}}(1),$$

which can be interpreted as uniform consistency of the “oracle” objective function (i.e., when $\hat{\omega}_i = \omega_i$).

3. Show that $\sup_{\theta \in \Theta} \left| \frac{1}{n} \sum_{i=1}^n \frac{\rho(\mathbf{z}_i; \theta)}{\hat{\omega}_i} - \mathbb{E} \left[\frac{\rho(\mathbf{z}_i; \theta)}{\omega_i} \right] \right| = o_{\mathbb{P}}(1)$.

Solution: We have

$$\begin{aligned}
& \sup_{\theta \in \Theta} \left| \frac{1}{n} \sum_{i=1}^n \frac{\rho(\mathbf{z}_i; \theta)}{\hat{\omega}_i} - \mathbb{E} \left[\frac{\rho(\mathbf{z}_i; \theta)}{\omega_i} \right] \right| \\
& \leq \sup_{\theta \in \Theta} \left| \frac{1}{n} \sum_{i=1}^n \frac{\rho(\mathbf{z}_i; \theta)}{\hat{\omega}_i} - \frac{1}{n} \sum_{i=1}^n \frac{\rho(\mathbf{z}_i; \theta)}{\omega_i} \right| + \sup_{\theta \in \Theta} \left| \frac{1}{n} \sum_{i=1}^n \frac{\rho(\mathbf{z}_i; \theta)}{\omega_i} - \mathbb{E} \left[\frac{\rho(\mathbf{z}_i; \theta)}{\omega_i} \right] \right| \\
& \leq \sup_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n \frac{|\hat{\omega}_i - \omega_i|}{\omega_i |\hat{\omega}_i|} |\rho(\mathbf{z}_i; \theta)| + o_{\mathbb{P}}(1) \\
& \leq \max_{1 \leq i \leq n} \frac{|\hat{\omega}_i - \omega_i|}{|\omega_i| |\hat{\omega}_i|} \frac{1}{n} \sum_{i=1}^n \sup_{\theta \in \Theta} |\rho(\mathbf{z}_i; \theta)| + o_{\mathbb{P}}(1) = o_{\mathbb{P}}(1),
\end{aligned}$$

because $\max_{1 \leq i \leq n} |\hat{\omega}_i - \omega_i| = o_{\mathbb{P}}(1)$ and $\inf_{1 \leq i \leq n} |\omega_i| \geq c$, which implies that $\inf_{1 \leq i \leq n} |\hat{\omega}_i| \geq c$ with probability approaching one, and $\frac{1}{n} \sum_{i=1}^n \sup_{\theta \in \Theta} |\rho(\mathbf{z}_i; \theta)| = \mathbb{E}[\sup_{\theta \in \Theta} |\rho(\mathbf{z}_i; \theta)|] + o_{\mathbb{P}}(1) = O_{\mathbb{P}}(1)$ by the law of large numbers.

4. Using part 2 above, show that $\hat{\theta}_n$ is a consistent estimator of θ .

Solution: Follows from the main result above upon setting

$$M_n(\theta) = \frac{1}{n} \sum_{i=1}^n \frac{\rho(\mathbf{z}_i; \theta)}{\hat{\omega}_i} \quad \text{and} \quad M(\theta) = \mathbb{E} \left[\frac{\rho(\mathbf{z}_i; \theta)}{\omega_i} \right],$$

because $\sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| = o_{\mathbb{P}}(1)$ was verified in the previous part, and the local identification condition holds by $\theta \mapsto M(\theta)$ being strictly convex.

The general consistency result above only requires uniform consistency of the sample objective function to the population objective function, and a local identification condition. It holds for estimators in arbitrary metric spaces, and it does not explicitly require a sample average structure or assumptions on the data dependence. The example above imposed further restrictions via the structure of objective function and the random sampling assumption, but still imposed a uniform consistency condition.

Therefore, it is natural to ask what additional general conditions guarantee uniform consistency of the sample objective function to the population objective function. To answer this question, assume the following four additional conditions:

- (Θ, d) is compact.
- $|M_n(\theta) - M(\theta)| = o_{\mathbb{P}}(1)$ for each $\theta \in \Theta$.
- $\theta \mapsto M(\theta)$ is continuous.

- $\theta \mapsto M_n(\theta)$ is *stochastic equicontinuous*: for any $\varepsilon, \eta > 0$ there exists $\delta = \delta(\varepsilon, \eta)$ (not depending on n) and a positive integer $n_0 = n_0(\varepsilon, \eta)$ such that for all $n \geq n_0$

$$\mathbb{P}\left(\sup_{\theta_1, \theta_2: d(\theta_1, \theta_2) < \delta} |M_n(\theta_1) - M_n(\theta_2)| > \eta\right) < \varepsilon.$$

The following questions employ the above assumptions.

5. By compactness, for each $\delta > 0$, there exist $t_1, t_2, \dots, t_M$ such that $\Theta \subseteq \bigcup_{j=1}^J B_j(\delta)$ with $B_j(\delta) = \{\theta \in \Theta : d(\theta, t_j) < \delta\}$, and $J = J(\delta)$ finite. Show that, for each $\delta > 0$,

$$\begin{aligned} & \sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| \\ & \leq \sup_{d(\theta_1, \theta_2) < \delta} |M_n(\theta_1) - M_n(\theta_2)| + \max_{1 \leq j \leq J} |M_n(t_j) - M(t_j)| + \sup_{d(\theta_1, \theta_2) < \delta} |M(\theta_1) - M(\theta_2)| \end{aligned}$$

Solution: We have

$$\begin{aligned} \sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| &= \max_{1 \leq j \leq J} \sup_{\theta \in B_j(\delta)} |M_n(\theta) - M(\theta)| \\ &= \max_{1 \leq j \leq J} \sup_{\theta \in B_j(\delta)} |M_n(\theta) - M_n(t_j) + M_n(t_j) - M(t_j) + M(t_j) - M(\theta)| \\ &\leq \max_{1 \leq j \leq J} \sup_{\theta \in B_j(\delta)} |M_n(\theta) - M_n(t_j)| + \max_{1 \leq j \leq J} |M_n(t_j) - M(t_j)| \\ &\quad + \max_{1 \leq j \leq J} \sup_{\theta \in B_j(\delta)} |M(t_j) - M(\theta)| \end{aligned}$$

and the result follows.

6. Show that for each $\eta > 0$ there exists $\delta > 0$ such that

$$\sup_{d(\theta_1, \theta_2) < \delta} |M(\theta_1) - M(\theta_2)| < \eta/3.$$

Solution: Because Θ is compact and M is continuous, it follows that M is uniform continuous. Thus, for any η , we can always find δ small enough so that the result holds.

7. Fixing $\varepsilon, \eta > 0$, show that $\mathbb{P}(\sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| > \eta) < \varepsilon$ for all large enough n , and explain why this implies uniform consistency $\sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| = o_{\mathbb{P}}(1)$.

Hint: choose $\delta = \min(\delta_1, \delta_2)$, where $\delta_1 = \delta_1(\varepsilon/2, \eta/3)$ is given by the definition of stochastic equicontinuity, and $\delta_2 = \delta_2(\eta)$ is given in part 6.

Solution: Fix $\varepsilon, \eta > 0$. By stochastic equicontinuity, there exists $\delta > 0$ such that

$$\limsup_{n \rightarrow \infty} \mathbb{P}\left(\sup_{\theta_1, \theta_2: d(\theta_1, \theta_2) < \delta} |M_n(\theta_1) - M_n(\theta_2)| > \eta/3\right) < \varepsilon/2.$$

By $|M_n(\theta) - M(\theta)| = o_{\mathbb{P}}(1)$ for each $\theta \in \Theta$, and using the union bound,

$$\begin{aligned} & \mathbb{P}\left(\max_{1 \leq j \leq J} |M_n(t_j) - M(t_j)| > \eta/3\right) \\ & \leq J \max_{1 \leq j \leq J} \mathbb{P}\left(|M_n(t_j) - M(t_j)| > \eta/3\right) < \varepsilon/2 \end{aligned}$$

for large enough n .

Then, using the above results, we conclude that for large enough n

$$\begin{aligned} & \mathbb{P}\left(\sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| > \eta\right) \\ & \leq \mathbb{P}\left(\sup_{d(\theta_1, \theta_2) < \delta} |M_n(\theta_1) - M_n(\theta_2)| > \eta/3\right) + \mathbb{P}\left(\max_{1 \leq j \leq J} |M_n(t_j) - M(t_j)| > \eta/3\right) < \varepsilon. \end{aligned}$$

Because ε is arbitrary, this means that for any $\eta > 0$ we have $\lim_{n \rightarrow \infty} \mathbb{P}(\sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| > \eta) = 0$.

Table 1.1. Discrete Distributions on $\mathcal{R}$

Uniform	p.d.f.	$1/m, x = a_1, \dots, a_m$
	m.g.f.	$\sum_{j=1}^m e^{a_j t}/m, t \in \mathcal{R}$
$DU(a_1, \dots, a_m)$	Expectation	$\sum_{j=1}^m a_j/m$
	Variance	$\sum_{j=1}^m (a_j - \bar{a})^2/m, \bar{a} = \sum_{j=1}^m a_j/m$
	Parameter	$a_i \in \mathcal{R}, m = 1, 2, \dots$
Binomial	p.d.f.	$\binom{n}{x} p^x (1-p)^{n-x}, x = 0, 1, \dots, n$
	m.g.f.	$(pe^t + 1 - p)^n, t \in \mathcal{R}$
$Bi(p, n)$	Expectation	np
	Variance	$np(1-p)$
	Parameter	$p \in [0, 1], n = 1, 2, \dots$
Poisson	p.d.f.	$\theta^x e^{-\theta}/x!, x = 0, 1, 2, \dots$
	m.g.f.	$e^{\theta(e^t - 1)}, t \in \mathcal{R}$
$P(\theta)$	Expectation	θ
	Variance	θ
	Parameter	$\theta > 0$
Geometric	p.d.f.	$(1-p)^{x-1}p, x = 1, 2, \dots$
	m.g.f.	$pe^t/[1 - (1-p)e^t], t < -\log(1-p)$
$G(p)$	Expectation	$1/p$
	Variance	$(1-p)/p^2$
	Parameter	$p \in [0, 1]$
Hyper-geometric	p.d.f.	$\binom{n}{x} \binom{m}{r-x} / \binom{N}{r}$
		$x = 0, 1, \dots, \min\{r, n\}, r - x \leq m$
$HG(r, n, m)$	m.g.f.	No explicit form
	Expectation	rn/N
	Variance	$rn m(N-r)/[N^2(N-1)]$
	Parameter	$r, n, m = 1, 2, \dots, N = n + m$
Negative binomial	p.d.f.	$\binom{x-1}{r-1} p^r (1-p)^{x-r}, x = r, r+1, \dots$
	m.g.f.	$p^r e^{rt}/[1 - (1-p)e^t]^r, t < -\log(1-p)$
$NB(p, r)$	Expectation	r/p
	Variance	$r(1-p)/p^2$
	Parameter	$p \in [0, 1], r = 1, 2, \dots$
Log-distribution	p.d.f.	$-(\log p)^{-1} x^{-1} (1-p)^x, x = 1, 2, \dots$
	m.g.f.	$\log[1 - (1-p)e^t]/\log p, t \in \mathcal{R}$
$L(p)$	Expectation	$-(1-p)/(p \log p)$
	Variance	$-(1-p)[1 + (1-p)/\log p]/(p^2 \log p)$
	Parameter	$p \in (0, 1)$

All p.d.f.'s are w.r.t. counting measure.

Table 1.2. Distributions on $\mathcal{R}$ with Lebesgue p.d.f.'s

Uniform	p.d.f.	$(b-a)^{-1}I_{(a,b)}(x)$
	m.g.f.	$(e^{bt} - e^{at})/[(b-a)t], \quad t \in \mathcal{R}$
$U(a, b)$	Expectation	$(a+b)/2$
	Variance	$(b-a)^2/12$
	Parameter	$a, b \in \mathcal{R}, \quad a < b$
Normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}$
	m.g.f.	$e^{\mu t + \sigma^2 t^2/2}, \quad t \in \mathcal{R}$
$N(\mu, \sigma^2)$	Expectation	μ
	Variance	σ^2
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$
Exponential	p.d.f.	$\theta^{-1} e^{-(x-a)/\theta} I_{(a, \infty)}(x)$
	m.g.f.	$e^{at} (1 - \theta t)^{-1}, \quad t < \theta^{-1}$
$E(a, \theta)$	Expectation	$\theta + a$
	Variance	θ^2
	Parameter	$\theta > 0, \quad a \in \mathcal{R}$
Chi-square	p.d.f.	$\frac{1}{\Gamma(k/2)2^{k/2}} x^{k/2-1} e^{-x/2} I_{(0, \infty)}(x)$
	m.g.f.	$(1 - 2t)^{-k/2}, \quad t < 1/2$
χ_k^2	Expectation	k
	Variance	$2k$
	Parameter	$k = 1, 2, \dots$
Gamma	p.d.f.	$\frac{1}{\Gamma(\alpha)\gamma^\alpha} x^{\alpha-1} e^{-x/\gamma} I_{(0, \infty)}(x)$
	m.g.f.	$(1 - \gamma t)^{-\alpha}, \quad t < \gamma^{-1}$
$\Gamma(\alpha, \gamma)$	Expectation	$\alpha\gamma$
	Variance	$\alpha\gamma^2$
	Parameter	$\gamma > 0, \quad \alpha > 0$
Beta	p.d.f.	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1} I_{(0,1)}(x)$
	m.g.f.	No explicit form
$B(\alpha, \beta)$	Expectation	$\alpha/(\alpha + \beta)$
	Variance	$\alpha\beta/[(\alpha + \beta + 1)(\alpha + \beta)^2]$
	Parameter	$\alpha > 0, \quad \beta > 0$
Cauchy	p.d.f.	$\frac{1}{\pi\sigma} \left[1 + \left(\frac{x-\mu}{\sigma} \right)^2 \right]^{-1}$
	ch.f.	$e^{\sqrt{-1}\mu t - \sigma t }$
$C(\mu, \sigma)$	Expectation	Does not exist
	Variance	Does not exist
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$

Table 1.2. (continued)

t-distribution	p.d.f.	$\frac{\Gamma[(n+1)/2]}{\sqrt{n\pi}\Gamma(n/2)} \left(1 + \frac{x^2}{n}\right)^{-(n+1)/2}$
	ch.f.	No explicit form
t_n	Expectation	0, ($n > 1$)
	Variance	$n/(n-2)$, ($n > 2$)
	Parameter	$n = 1, 2, \dots$
F-distribution	p.d.f.	$\frac{n^{n/2}m^{m/2}\Gamma[(n+m)/2]x^{n/2-1}}{\Gamma(n/2)\Gamma(m/2)(m+nx)^{(n+m)/2}}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$F_{n,m}$	Expectation	$m/(m-2)$, ($m > 2$)
	Variance	$2m^2(n+m-2)/[n(m-2)^2(m-4)]$, ($m > 4$)
	Parameter	$n = 1, 2, \dots$, $m = 1, 2, \dots$
Log-normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}x^{-1}e^{-(\log x - \mu)^2/2\sigma^2}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$LN(\mu, \sigma^2)$	Expectation	$e^{\mu+\sigma^2/2}$
	Variance	$e^{2\mu+\sigma^2}(e^{\sigma^2} - 1)$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$
Weibull	p.d.f.	$\frac{\alpha}{\theta}x^{\alpha-1}e^{-x^\alpha/\theta}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$W(\alpha, \theta)$	Expectation	$\theta^{1/\alpha}\Gamma(\alpha^{-1} + 1)$
	Variance	$\theta^{2/\alpha} \left\{ \Gamma(2\alpha^{-1} + 1) - [\Gamma(\alpha^{-1} + 1)]^2 \right\}$
	Parameter	$\theta > 0$, $\alpha > 0$
Double Exponential	p.d.f.	$\frac{1}{2\theta}e^{- x-\mu /\theta}$
	m.g.f.	$e^{\mu t}/(1 - \theta^2 t^2)$, $ t < \theta^{-1}$
$DE(\mu, \theta)$	Expectation	μ
	Variance	$2\theta^2$
	Parameter	$\mu \in \mathcal{R}$, $\theta > 0$
Pareto	p.d.f.	$\theta a^\theta x^{-(\theta+1)}I_{(a,\infty)}(x)$
	ch.f.	No explicit form
$Pa(a, \theta)$	Expectation	$\theta a/(\theta - 1)$, ($\theta > 1$)
	Variance	$\theta a^2/[(\theta - 1)^2(\theta - 2)]$, ($\theta > 2$)
	Parameter	$\theta > 0$, $a > 0$
Logistic	p.d.f.	$\sigma^{-1}e^{-(x-\mu)/\sigma}/[1 + e^{-(x-\mu)/\sigma}]^2$
	m.g.f.	$e^{\mu t}\Gamma(1 + \sigma t)\Gamma(1 - \sigma t)$, $ t < \sigma^{-1}$
$LG(\mu, \sigma)$	Expectation	μ
	Variance	$\sigma^2\pi^2/3$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$